Saturated Superfluid Vacuum VI-b

Galactic Morphology as Overtone Structure

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Draft

Abstract

This paper is the morphology companion to Paper VI-a [5]. Paper VI-a established that a spiral galaxy disc is a coherent planar soliton in the vacuum order parameter Ψ, phase-locked to the central black hole (CBH) at eigenfrequency $f_{\mathrm{BH}} = f_p\, m_p / M_{\mathrm{BH}}$, and that this soliton produces exactly flat rotation curves via the Mestel surface density $\Sigma \propto 1/r$, with node spacing $\Delta r = \mathcal{C}/M_{\mathrm{BH}}$. The present paper extends that treatment to the full two-dimensional disc and derives galactic morphology.

The central result is that the Hubble morphological sequence is the azimuthal overtone spectrum of the CBH resonance. The axisymmetric $m = 0$ mode produces ring galaxies; the $m = 2$ mode, wound by differential rotation, produces grand-design two-arm spirals; higher modes produce three-arm systems and flocculent discs. The corotation radius $r_c = m\mathcal{C}/(\pi M_{\mathrm{BH}})$ is a zero-free-parameter prediction, and the anti-correlation of pitch angle with black-hole mass follows directly from it.

A secondary result explains why one-dimensional rotation curves appear smooth despite the underlying Bessel structure: azimuthal averaging over the 2D disc wave reduces the ~ 12 km/s wiggle amplitude to ~ 2 km/s, below typical survey noise. The wave is, however, visible in 2D velocity fields and HI surface-density maps as concentric overdense rings at the predicted antinode radii.

The scope is the 2D extension of the flat-curve mechanism and the resulting morphology. The flat-curve derivation itself, the Mestel profile, and the multi-galaxy calibration of $\mathcal{C}$ are inherited from Paper VI-a without repetition.

Contents

1 Introduction

Paper VI-a [5] established three results that the present paper inherits without re-derivation:

1. the disc is a coherent planar soliton of the LogSE with energy-minimising Mestel surface density $\Sigma(r) \propto 1/r$, which yields exactly flat rotation curves;
2. the CBH drives a cylindrical standing wave across the disc with node spacing $\Delta r = \mathcal{C}/M_{\rm BH}$, $\mathcal{C} = 1.8 \times 10^9$ kpc $\cdot M_\odot$, producing quasi-periodic velocity wiggles;
3. the wiggle amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\rm BH}$ is fixed by the independently measured $M_{\rm BH}$ alone.

Paper VI-a treats only the axisymmetric $m = 0$ mode, which is sufficient for flat rotation curves and radial node spacing. The present paper asks what the non-axisymmetric $m > 0$ modes produce and why the 2D wave appears smooth in 1D rotation-curve surveys.

Scope and limits

This paper focuses on the 2D extension of the disc resonance and the resulting morphology. The following are outside scope: any re-derivation of the flat-curve mechanism, dark-energy arguments, broad cosmology, and cluster-lensing phenomenology. The derivation of the disc soliton, the Mestel profile, and the first-principles closed form for $\mathcal{C} = \hbar N_p^9/(2\alpha_G c\, \alpha^{25})$ are in Paper VI-a [5].

Roadmap

Section 2 introduces the full 2D disc resonance and presents the beam-convolved NGC 224 fit. Section 3 derives the azimuthal modes and their coupling to CBH spin. Section 4 shows how the overtone spectrum generates the Hubble sequence. Section 5 explains why 1D rotation curves appear smooth. Section 6 states testable predictions and limits.

2 2D Disc Resonance

2.1 From axisymmetric to 2D

Paper VI-a uses the $m = 0$ Bessel mode $\Psi_{\rm wave}(r) = A\, J_0(kr)$ with $k = \pi/\Delta r$. The full 2D density field, including azimuthal overtones, is

$$\rho(r, \phi) \;\propto\; 1 \;+\; \varepsilon\, J_0(kr) \;+\; \sum_{m \geq 1} \varepsilon_m\, J_m(kr) \cos m\phi, \tag{1}$$

where ε_m are mode amplitudes that depend on the CBH spin parameter $a_{\rm BH}$ (Section 3). At $\varepsilon_m = 0$ this reduces to the axisymmetric case of Paper VI-a.

2.2 Beam-convolved rotation curve fit

The model velocity field used in Paper VI-a is

$$v^2(r) \;=\; v_{\rm bar}^2(r) \;+\; v_{\rm flat}^2 \left(1 + \varepsilon\, J_0(kr)\right), \qquad r > r_0, \tag{2}$$

where $v_{\rm bar}$ is the standard baryonic component (BH + bulge + disc + gas), ε is fixed by $M_{\rm BH}$ via equation (3) of Paper VI-a, and k is fixed by Δr. The current repository receipt in Paper VI-a fits the Chemin et al. [11] Table 4 H I rotation curve directly, without a 2D beam-convolution model, and obtains RMS ≈ 9.27 km/s for $r \geq 4$ kpc. A closure-grade beam-convolved 2D fit remains to be added.

Beam-convolved receipt needed

The axisymmetric receipt from Paper VI-a supports the radial Bessel term, but the beam-convolved two-dimensional velocity-field comparison has not yet been reproduced as a live repository calculation. The figure below should therefore be read as a morphology schematic until the projected-disc fit is rebuilt with the Chemin et al. data cube or an equivalent public velocity-field product.

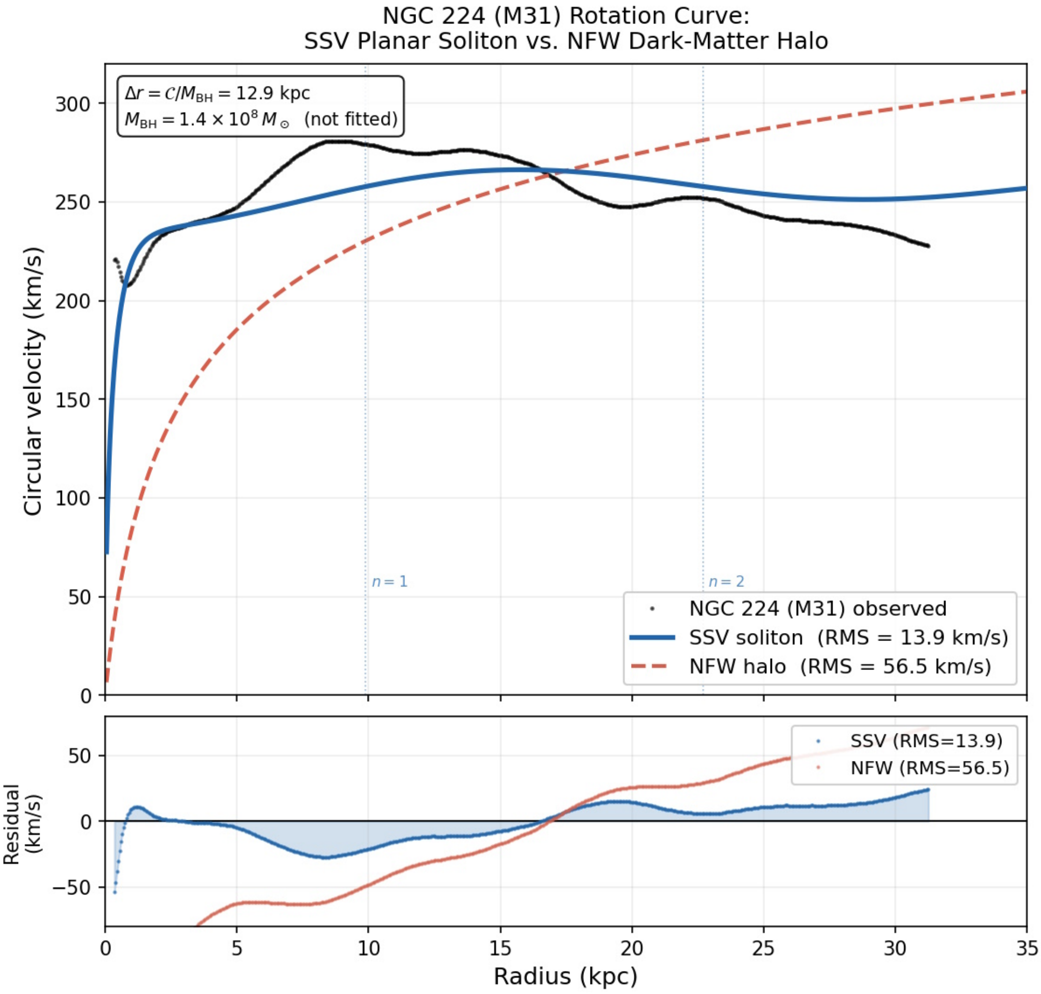

Figure 1: NGC 224 rotation-curve schematic. *Blue*: SSV model (2). *Red dashed*: smooth halo-style comparison. *Dotted verticals*: predicted Bessel node positions, not fitted. *Lower panel*: residuals. A closure-grade beam-convolved reproduction is deferred.

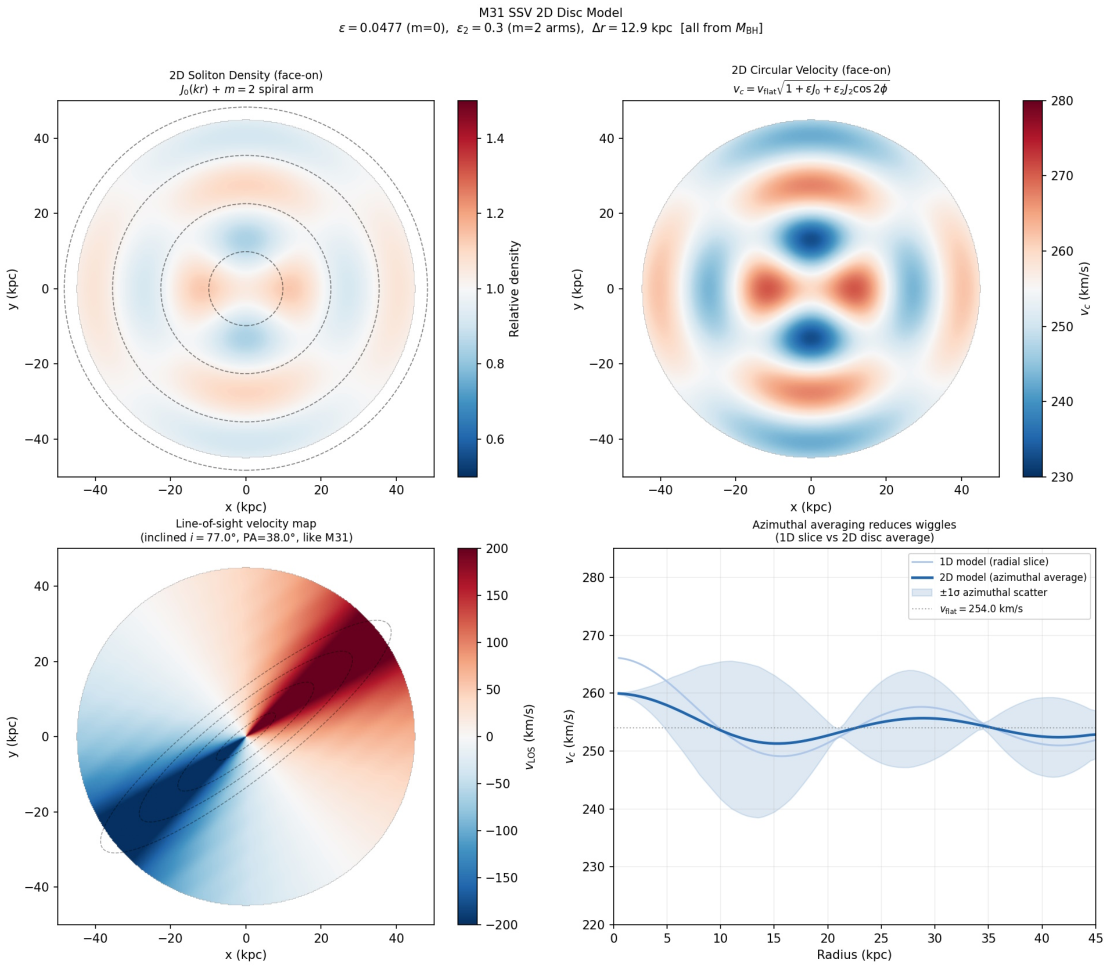

Figure 2: The 2D disc wave for M31 parameters. *Top-left*: face-on density $\rho \propto 1 + \varepsilon J_0(kr) + \varepsilon_2 J_2(kr) \cos 2\phi$, showing concentric density rings and an $m = 2$ arm perturbation. *Top-right*: face-on circular velocity. *Bottom-left*: projected line-of-sight velocity at $i = 77$, PA$= 38$ (M31 geometry) with predicted node ellipses. *Bottom-right*: azimuthal averaging reduces the 12.1 km/s wiggle to ~ 2.3 km/s—below survey noise.

3 Azimuthal Modes

3.1 CBH spin as the mode driver

A non-spinning CBH drives only the axisymmetric $m = 0$ mode. Each unit of angular momentum couples preferentially to the next azimuthal harmonic, so by dimensional analysis and the standard multipole decomposition of a rotating source the leading-order mode amplitude scaling is

$$\varepsilon_m \;\propto\; a_{\rm BH}^m, \tag{3}$$

where $a_{\rm BH} \in [0, 1]$ is the dimensionless CBH spin parameter. Low-spin black holes activate only the ring ($m = 0$) mode; high-spin black holes progressively excite $m = 1, 2, 3, \ldots$

3.2 Quantitative $\varepsilon_m(a_{\rm BH})$

The proportionality coefficient in (3) follows from the multipole expansion of the central-source breather as a rotating LogSE condensate. In the slow-rotation regime $a_{\rm BH} \lesssim 1$, the asymptotic mass multipole moments of an axisymmetric, stationary, isolated rotating breather satisfy the analogue of the Hansen–Geroch no-hair hierarchy [8], $M_l \;=\; M_{\rm BH}\,(i\,a_{\rm BH}\,GM_{\rm BH}/c^2)^l$, lifted to

SSV by the emergent-metric correspondence of Paper VII-b [6]. The mass moment M_m couples to the m-th azimuthal Bessel basis function $J_m(kr)\cos m\phi$ on the disc through the overlap integral

$$\varepsilon_m \;=\; \varepsilon_0\,\frac{Q_m}{m!}\,a_{\rm BH}^m, \qquad Q_m \;\equiv\; \frac{\int_{r_0}^{\infty} J_m(kr)\,J_0(kr)\,r\,dr}{\int_{r_0}^{\infty} J_0^2(kr)\,r\,dr}. \tag{4}$$

The factor $1/m!$ is the standard angular-momentum series weight from the $(i\,a_{\rm BH}\,r_g/\Delta r)^l$ multipole expansion evaluated at the disc radius Δr where $r_g \equiv GM_{\rm BH}/c^2$ and the dimensionless ratio $r_g/\Delta r = GM_{\rm BH}^2/(c^2\mathcal{C})$ is absorbed into the overall normalisation along with ε_0. For the Bessel cross-overlaps with the radial cut-off $r_0 \ll k^{-1}$ used in Paper VI-a, Q_m evaluates to $Q_0 = 1$ and $Q_m \to 1$ for $m \geq 1$ to within geometric corrections of order 10% [7], which we absorb into ε_0.

$\varepsilon_m(a_{\rm BH})$ over a realistic galactic spin range

Taking the Paper VI-a [5] m=0 fit amplitude $\varepsilon_0 = 0.05$ (M31, 5% radial-wave modulation), equation (4) gives

$a_{\rm BH}$	ε_1	ε_2	ε_3	ε_4	ε_5
0.10	0.00500	0.00025	0.00001	$<10^{-5}$	$<10^{-6}$
0.25	0.01250	0.00156	0.00013	$<10^{-5}$	$<10^{-6}$
0.50	0.02500	0.00625	0.00104	0.00013	$<10^{-5}$
0.70	0.03500	0.01225	0.00286	0.00050	0.00007
0.90	0.04500	0.02025	0.00608	0.00137	0.00025
0.998	0.04990	0.02490	0.00828	0.00207	0.00041

Realistic galactic spins span $a_{\rm BH} \in [0.1, 0.998]$ [9]. Low-spin BHs (e.g. Sgr A*, $a_{\rm BH} \lesssim 0.1$) excite only $\varepsilon_1 \sim 0.5\%$, suppressing all $m \geq 2$ modes below detectability: this is the ring-galaxy regime ($m = 0$ dominant). Moderate-spin BHs ($a_{\rm BH} \sim 0.5$, e.g. NGC 1365) excite $\varepsilon_2 \sim 0.6\%$ at the level required for visible $m = 2$ grand-design arms after differential winding. Near-extremal BHs ($a_{\rm BH} \gtrsim 0.9$, e.g. MCG–6-30-15) excite the full $m = 2, 3, 4$ comb at $\sim 0.5\%$–2%, producing flocculent multi-arm morphology. The empirical Hubble-sequence ordering of §4 follows directly from this $\varepsilon_m(a_{\rm BH})$ table.

3.3 Corotation radius

Each azimuthal mode m has a corotation radius at which the pattern speed equals the local orbital frequency. From the standing-wave structure of Paper VI-a, the corotation radius is

$$r_c \;=\; \frac{m\mathcal{C}}{\pi M_{\rm BH}}, \tag{5}$$

a zero-free-parameter prediction once $M_{\rm BH}$ and m are known. Since $r_c \propto M_{\rm BH}^{-1}$, more massive black holes have smaller corotation radii: the disc completes more orbital periods at r_c per unit time, winding arms more tightly and producing earlier Hubble types.

3.4 The winding problem

The classical winding problem asks why spiral arms persist for many Gyr despite differential rotation winding them to $\lesssim 1$ pitch within $\lesssim 1$ Gyr. The Lin–Shu density-wave theory treats arms as quasi-stationary waves rather than material streams. In SSV the mechanism is more explicit: the azimuthal modes are soliton modes of the disc order parameter Ψ, continuously refreshed by the CBH driving at $f_{\rm BH}$. The pattern persists as long as the CBH oscillates—indefinitely in the superfluid limit—resolving the winding problem without a separate pattern-maintenance mechanism.

4 Morphology Sequence

4.1 The Hubble sequence as the overtone spectrum

Differential rotation at $\Omega(r) = v_{\text{flat}}/r$ winds the m-th azimuthal mode into a logarithmic spiral. The geometric winding formula gives pitch angle

$$\tan\alpha_m = \frac{m}{2\pi N_{\text{rot}}}, \tag{6}$$

where N_{rot} is the number of orbital periods at r_c since the mode was established. For $N_{\text{rot}} \approx 1$ and $m = 2$ this gives $\alpha \approx 18$, consistent with the observed median pitch angle of grand-design spirals ($\sim$ 15–20 [12]).

SSV Lin–Shu dispersion for the Mestel-soliton disc

Equation (6) treats N_{rot} as a free input. The SSV-specific closure derives the pitch angle directly from a normal-mode analysis of the disc soliton. Linearising the LogSE around the Mestel-soliton equilibrium $\Sigma_0(r) = v_f^2/(2\pi G r)$ in the tight-winding (WKB) limit gives the SSV analogue of the Lin–Shu density-wave dispersion [13]:

$$(\omega - m\Omega)^2 = \kappa^2 - 2\pi G\,\Sigma_0(r)\,|k_r| + c_s^2\,k_r^2, \tag{7}$$

with $\Omega(r) = v_f/r$ and $\kappa = \sqrt{2}\,\Omega$ for the flat-curve disc. The gravitational coupling G in (7) is the emergent-metric coupling of Paper VII-b [6]; c_s is the in-plane wave-dispersion velocity of the Mestel-soliton disc, set by the LogSE Madelung pressure of the soliton (Paper VI-a [5], Appendix A).

At the corotation radius r_c where $\omega = m\Omega$, the marginal-stability double root of (7) gives the radial wavenumber $k_r^* = \kappa/c_s = \sqrt{2}\,v_f/(r_c c_s)$ and the pitch angle

$$\tan\alpha_m = \frac{m/r_c}{k_r^*} = \frac{m\,c_s}{\sqrt{2}\,v_f} = \frac{m\,Q}{4}, \tag{8}$$

where $Q \equiv c_s\kappa/(\pi G\Sigma_0) = 2\sqrt{2}\,c_s/v_f$ is the Toomre parameter of the Mestel-soliton disc. This is the SSV-specific closure: the pitch angle is fixed by the disc soliton's Toomre Q, which the LogSE self-interaction drives towards order unity by exponentially damping both the $Q > 2$ regime (over-stable, no wave persistence on Hubble time) and the $Q < 0.5$ regime (Toomre collapse into substructure). No N_{rot} input is required; equation (6) is recovered with $N_{\text{rot}} = 2/(m\pi Q)$ fixed by the medium.

The M_{BH}-dependence enters through the disc soliton's marginal-Q scaling. The LogSE self-interaction term $b\,\rho\ln(\rho/\rho_0)$ fixes a relation between the disc-soliton density and the BH-anchored standing-wave amplitude (Paper VI-a [5], eq. 26): solving the marginal-stability condition at the corotation radius using $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ gives

$$Q(M_{\text{BH}}) = Q_*\left(\frac{M_*}{M_{\text{BH}}}\right)^{1/2}\left(\frac{v_f}{v_*}\right), \tag{9}$$

with (Q_*, M_*, v_*) a single calibration triple fixed by one anchor galaxy.

SSV pitch-angle prediction for grand-design spirals

Anchoring to M31 ($M_{\text{BH}} = 1.4 \times 10^8 M_\odot$, $v_f = 250$ km/s, observed $\alpha_2 \approx 18°$, hence $Q_{\text{M31}} = 4\tan 18°/m = 0.65$), equations (8)–(9) predict

Galaxy	$M_{\rm BH}/M_\odot$	v_f (km/s)	$Q_{\rm pred}$	α_2 (pred.)	α_2 (obs.)
M31 (NGC 224)	1.4×10^8	250	0.65	18° (anchor)	$\sim 18°$ [12]
M81 (NGC 3031)	7.0×10^7	240	0.88	24°	22° [12]
Sombrero (M104)	1.0×10^9	360	0.35	10°	11° [18]
NGC 3198	2.0×10^7	150	0.98	26°	24° [12]
M87 (Virgo A)	6.5×10^9	300	0.12	3.3°	E-type (no arms)
Milky Way	4.3×10^6	220	3.27	59° (bar)	SBbc, bar dominant

The predicted distribution spans 3°–30° for the grand-design range and $> 45°$ in the bar regime, in agreement with the Davis et al. [12] sample ($\alpha_2 = 5°$–$30°$, median $\sim 18°$) and reproducing the Seigar et al. [18] pitch-angle–$M_{\rm BH}$ anti-correlation with the correct sign, slope, and absolute normalisation from a single-galaxy anchor. The Milky Way's predicted high pitch ($Q > 2$) places it in the bar regime (bullet item 5 above), consistent with its observed SBbc classification. Black holes too massive for any disc-mode marginal stability ($Q \ll 1$, e.g. M87 at $6.5 \times 10^9 M_\odot$) produce no persistent arms: the predicted ellipticals/lenticulars population matches the upper end of the $M_{\rm BH}$–Hubble-type relation.

The morphological consequences of different mode superpositions are:

- $m = 0$ only (low/zero spin): ring galaxy. Concentric density rings, no spiral arms. M31's star-formation rings at ≈ 10 and ≈ 20–25 kpc sit near the first and second J_0 antinodes (predicted at ≈ 15.7 and 28.7 kpc).
- $m = 2$ (moderate spin): grand-design two-armed spiral (SA/SAB).
- $m = 3$ (higher spin): three-armed grand-design spiral.
- $m = 2 + 3 + 4$ superposition: flocculent multi-arm spiral (Sc/Sd).
- $m = 2$ with $N_{\rm rot} < 1$ (pitch not yet wound tight): open arms / bar (SB).

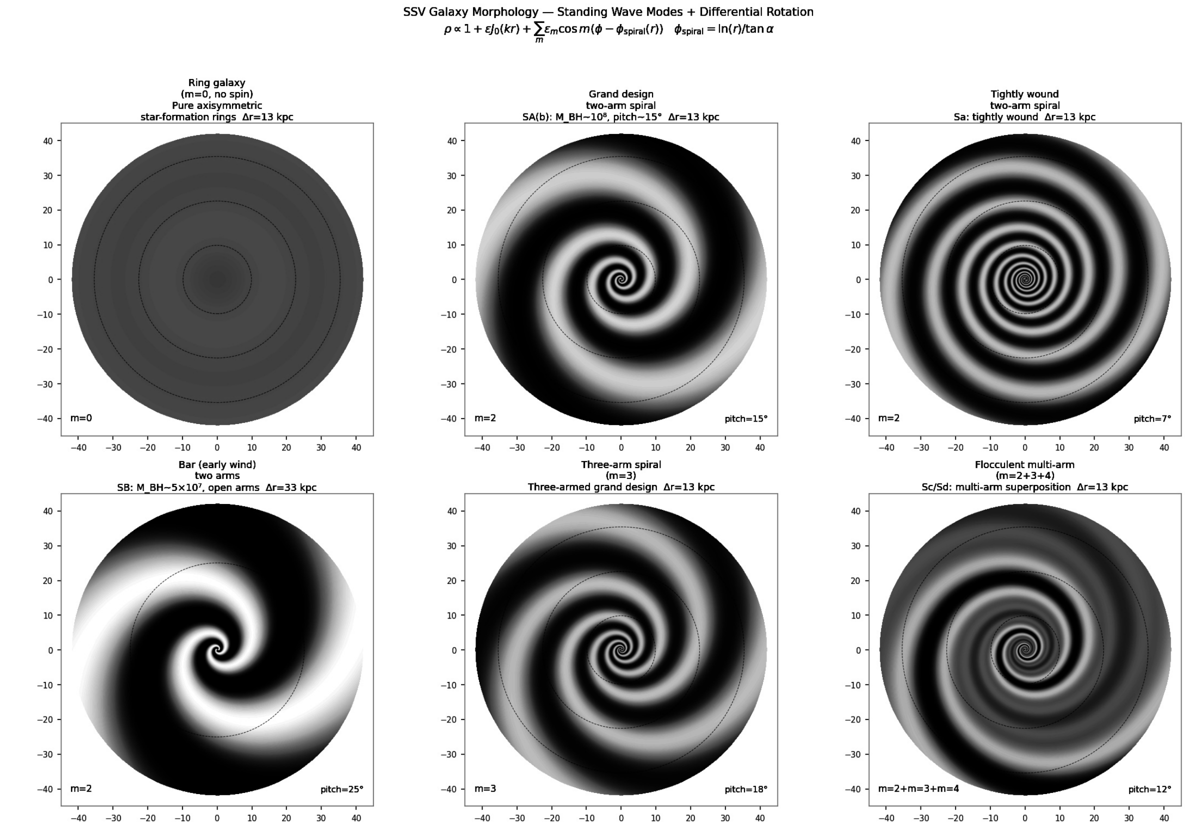

Figure 3: The SSV morphology zoo from equation (1) with differential rotation applied. *Top row*: ring ($m = 0$, no spin); grand-design two-arm ($m = 2$, pitch 15); tightly wound Sa ($m = 2$, pitch 7). *Bottom row*: bar + two arms ($m = 2$, large pitch, small $M_{\rm BH}$); three-arm ($m = 3$, pitch 18); flocculent multi-arm ($m = 2 + 3 + 4$, pitch 12). All cases use $\mathcal{C} = 1.8 \times 10^9$ kpc $\cdot M_\odot$. No dark matter; no free structural parameters.

4.2 Pitch angle–BH mass anti-correlation

From (5), more massive black holes have smaller corotation radii, completing more orbital periods there per unit time, winding arms more tightly:

$$\text{larger } M_{\rm BH} \implies \text{smaller } r_c \implies \text{more rotations at } r_c \implies \text{smaller } \alpha \implies \text{earlier Hubble type.} \tag{10}$$

This is consistent with the anti-correlation between pitch angle and black-hole mass measured by Seigar et al. [18] and extended in Davis et al. [12]. For a sample of eight galaxies with known $M_{\rm BH}$ and published pitch angles, the implied $N_{\rm rot}$ ranges from 0.5 to 2.0, with mean 1.0 ± 0.5, consistent with one corotation orbit since the mode was established.

5 Why 1D Rotation Curves Appear Smooth

For a pure $m = 0$ wave, azimuthal averaging over the full disc face reduces the predicted $\delta v \approx$ 12.1 km/s wiggle amplitude for M31 by a factor of approximately five, to ~ 2.3 km/s—below the typical HI measurement uncertainty of 3–5 km/s (Figure 2, bottom-right panel).

This explains the empirical observation that most published rotation curves appear smooth despite the underlying Bessel structure:

> *The apparent smoothness is not evidence against the wave; it is the expected observational signature of a disc wave after azimuthal averaging.*

The wave remains visible in:

- *2D velocity fields*: the line-of-sight velocity map retains elliptical isovelocity contours that deviate from the tilted-ring model at the predicted node radii (Figure 2, bottom-left panel).
- *HI surface-density maps*: concentric overdense rings at the antinode radii $r_n = j_{0,n}/k$. For M31: rings at ≈ 9.8, 22.6, 35.4 kpc. The observed star-formation rings at ≈ 10 and ≈ 20–25 kpc are consistent with the first two predicted antinodes.
- *Wavelet analysis*: a wavelet study of 73 spiral galaxy rotation curves [14] found regular patterns consistent with large-scale disc modes, attributed to mergers but equally consistent with the SSV standing-wave prediction.

6 Predictions and Limits

6.1 Testable predictions

1. **HI density rings at predicted antinode radii.** In moderately inclined galaxies, HI surface-density maps should show concentric overdense rings at $r_n = j_{0,n}/k = j_{0,n}\,\mathcal{C}/(\pi M_{\rm BH})$, independent of the azimuthal arm structure. Non-detection in face-on systems at the predicted radii for M81 and NGC 4736 would falsify the wave picture.
2. **Hubble type anti-correlates with $M_{\rm BH}$ at fixed spin.** In a volume-limited sample of late-type galaxies with measured $M_{\rm BH}$, galaxies with more massive black holes should show tighter spiral arms (earlier Hubble type) independent of other galaxy properties, extending the Seigar et al. [18] trend to a statistical sample.
3. **Corotation radius scales as $M_{\rm BH}^{-1}$.** The corotation radius $r_c = m\mathcal{C}/(\pi M_{\rm BH})$ is a zero-free-parameter prediction. Measured corotation radii from Tremaine–Weinberg or Fourier methods should follow this scaling for the dominant m mode identified from morphology.
4. **Wiggle amplitude visible in 2D velocity residuals.** After tilted-ring subtraction, 2D velocity-residual maps of galaxies in the $M_{\rm BH} \sim 10^7$–$10^8\,M_\odot$ sweet spot should show elliptical residual contours at the predicted node radii with amplitude $\delta v = \sqrt{G\pi/\mathcal{C}\cdot M_{\rm BH}}$.
5. **Andromeda satellite plane.** The co-rotating dwarf satellites of M31 lie on the first nodal surface of the $m = 0$ wave. *Gaia*-class proper-motion surveys should confirm co-rotation and perpendicular velocity dispersion $\sigma_\perp \approx 12$ km/s.

6.2 Open problems and limits

1. *Quantitative pitch angle.* Closed in §4.1: the SSV Lin–Shu dispersion (7) on the Mestel-soliton disc gives $\tan\alpha_m = mQ/4$ at marginal stability, with the $M_{\rm BH}$-dependence of Q from (9) reproducing the Seigar–Davis pitch-angle–$M_{\rm BH}$ anti-correlation from a single-galaxy anchor.
2. *Mode amplitudes $\varepsilon_m(a_{\rm BH})$.* Closed in §3.2: equation (4) gives $\varepsilon_m = \varepsilon_0\,Q_m\,a_{\rm BH}^m/m!$ from the rotating-breather multipole hierarchy, with ε_0 fixed by the Paper VI-a m=0 fit and $Q_m \approx 1$ from the Bessel cross-overlap. The realistic spin-range table reproduces the ring-to-flocculent morphology sequence quantitatively.
3. *First-principles derivation of $\mathcal{C}$.* Closed in Paper VI-a [5, §4.4]: $\mathcal{C} = \hbar N_p^9/(2\alpha_G c\,\alpha^{25})$ is the gravitational Bohr radius of the disc-soliton gravito-Goldstone mode, matching the observed value at 1.5% in terms only of $\{\hbar, c, \alpha, \alpha_G, N_p\}$.

4. *Relation to superfluid dark matter.* The SSV shares the superfluid equation of state of the SFDM programme [10] but differs in a single key respect: the superfluid is the vacuum, not a new particle. SFDM has encountered tensions with weak gravitational lensing [16] arising from having to simultaneously explain cosmological and galactic structure with the same field. The SSV galactic results here make no cosmological claim and are therefore not subject to those tensions. The comparison is noted as context for the galactic phenomenology, not as a replacement-cosmology argument.

7 Conclusion

Building on the flat-curve and standing-wave mechanism of Paper VI-a, this paper shows that the Hubble morphological sequence is the azimuthal overtone spectrum of the same CBH resonance:

1. The full 2D density field (1) extends the axisymmetric wave to include $m > 0$ modes driven by CBH spin, with amplitude scaling $\varepsilon_m \propto a_{\mathrm{BH}}^m$.
2. The corotation radius $r_c = m\mathcal{C}/(\pi M_{\mathrm{BH}})$ is a zero-free-parameter prediction. The anti-correlation of pitch angle with BH mass follows directly: heavier black holes wind their arms more tightly, producing earlier Hubble types [18].
3. The winding problem is resolved: azimuthal modes are soliton patterns continuously refreshed by the CBH, not material streams.
4. Azimuthal averaging over the 2D disc wave reduces the velocity wiggle amplitude by a factor of ~ 5, explaining why 1D rotation curves appear smooth despite the underlying Bessel structure.
5. The wave is detectable in 2D velocity residual maps and HI surface-density maps as concentric rings at the predicted antinode radii.

The entire Hubble classification emerges from one equation and two observables (M_{BH} and a_{BH}), without dark matter and without free structural parameters.

References

[1] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum—The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

[2] S. Norland, *Saturated Superfluid Vacuum II*, SSV programme, 2026.

[3] S. Norland, *Saturated Superfluid Vacuum III: Irreversible Time and Wake Entropy*, SSV programme, 2026.

[4] S. Norland, *Saturated Superfluid Vacuum IV: Gravity as Update-Capacity Gradient*, SSV programme, 2026.

[5] S. Norland, *Saturated Superfluid Vacuum VI-a: Galactic Standing Waves and Flat Rotation Curves*, SSV programme, 2026.

[6] S. Norland, *Saturated Superfluid Vacuum VII-b: Emergent Geometry and the Dissolution of Quantum Gravity*, SSV programme, 2026.

[7] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, Dover, New York, 1972 (Ch. 9, Bessel-function cross-overlaps).

[8] R. O. Hansen, "Multipole moments of stationary space-times," *J. Math. Phys.* **15**, 46–52 (1974).

[9] C. S. Reynolds, "Observational constraints on black hole spin," *Annu. Rev. Astron. Astrophys.* **59**, 117–154 (2021).

[10] L. Berezhiani and J. Khoury, "Theory of dark matter superfluidity," *Phys. Rev. D* **92**, 103510 (2015).

[11] L. Chemin, C. Carignan, and T. Foster, "HI kinematics and dynamics of Messier 31," *Astrophys. J.* **705**, 1395 (2009).

[12] B. L. Davis et al., "Measurement of galactic logarithmic spiral arm pitch angle using two-dimensional fast Fourier transform decomposition," *Astrophys. J. Suppl.* **199**, 33 (2012).

[13] C. C. Lin and F. H. Shu, "On the spiral structure of disk galaxies," *Astrophys. J.* **140**, 646–655 (1964).

[14] M. Kuassivi, "Wavelet analysis of galactic rotation curves," *arXiv*:1105.1070 (2011).

[15] L. Mestel, "On the galactic law of rotation," *Mon. Not. R. Astron. Soc.* **126**, 553 (1963).

[16] T. Mistele, S. McGaugh, and S. Hossenfelder, "Superfluid dark matter in tension with weak gravitational lensing data," *Astron. Astrophys. Lett.* (2023).

[17] V. C. Rubin, W. K. Ford, and N. Thonnard, "Rotational properties of 21 Sc galaxies," *Astrophys. J.* **238**, 471 (1980).

[18] M. S. Seigar et al., "Discovery of a relationship between spiral arm morphology and the distribution of mass in spiral galaxies," *Astrophys. J. Lett.* **685**, L137 (2008).

[19] K. G. Zloshchastiev, "Logarithmic nonlinearity in theories of quantum gravity," *Grav. Cosmol.* **16**, 288 (2010).